ON THE COMPATIBILITY OF MODEL-SPACE REPRESENTATIONS WITH THE FUNDAMENTAL THEOREM OF ARITHMETIC

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Abstract. We formalize the algebraic interface between a structured model space $\mathcal{M}$ and the classical set of natural numbers $\mathbb{N}$. Specifically, we address the apparent contradiction when a natural number $n \in \mathbb{N}$ admits both a semi-prime decomposition representation $a \star b \star E_1$ and a compact state representation $c \star E_2$ within $\mathcal{M}$. By strictly disentangling the unique prime factorization of the projected scalar $n = \pi(x)$ from the factorization theory of state elements $x \in \mathcal{M}$, we prove that scalar equivalence $\pi(x) = \pi(y)$ implies neither algebraic association ($x \sim y$) nor isomorphism of state-level factorizations ($\text{Fact}(x) = \text{Fact}(y)$). This provides a rigorous mathematical safeguard preserving the Fundamental Theorem of Arithmetic across non-trivial model spaces.

1. Introduction

In various structural and algebraic physics models, state spaces $\mathcal{M}$ are equipped with product operations $\star$ and special unit-like elements $E_i \in \mathcal{M}$. A recurring structural question concerns the validity of the Fundamental Theorem of Arithmetic when a given scalar quantity admits multiple algebraic representations.

Consider two states $x, y \in \mathcal{M}$ given by

$$x = a \star b \star E_1, \quad \text{and} \quad y = c \star E_2, \tag{1}$$

where a, b correspond to prime numbers in $\mathbb{N}$, E_1, E_2 act as sectorial boundary or phase operators, and $c \in \mathcal{M}$ is an irreducible element in the state space. If one naively equates $x = y$ within $\mathbb{N}$, an immediate contradiction with the uniqueness of prime factorizations arises, as c would simultaneously act as a new prime and as a composite scalar $a \cdot b$.

The goal of this paper is to construct a rigorous projection framework $\pi : \mathcal{M} \to \mathbb{N}$ that reconciles the algebraic richness of $\mathcal{M}$ with the absolute uniqueness of factorizations in $\mathbb{N}$.

2. Formal Framework and Projection Semantics

Let $\mathcal{M}$ be a non-empty set (the *model space*) endowed with a binary operation $\star : \mathcal{M} \times \mathcal{M} \to \mathcal{M}$.

Definition 2.1 (Scalar Projection Map)**.** A map $\pi : \mathcal{M} \to \mathbb{N}$ is called a *scalar projection*. For any state $x \in \mathcal{M}$, the image $n = \pi(x) \in \mathbb{N}$ is called the *projected scalar value* of x.

Remark 2.2 (Fiber Structure vs. Fundamental Theorem). The Hauptsatz (Fundamental Theorem of Arithmetic) applies strictly to the target object $n = \pi(x) \in \mathbb{N}$. It does not constrain the cardinality or structural complexity of the fiber (pre-image):

$$\pi^{-1}(n) = \{x \in \mathcal{M} \mid \pi(x) = n\}. \tag{2}$$

While n possesses a unique canonical prime factorization

$$n = \prod_{p \text{ prime}} p^{v_p(n)}, \quad v_p(n) \in \mathbb{N}_0, \tag{3}$$

the fiber $\pi^{-1}(n)$ may contain an arbitrarily large number of distinct model states $x \neq y$.

Definition 2.3 (Distinction between Projection and Norm)**.** A scalar projection $\pi : \mathcal{M} \to \mathbb{N}$ must be carefully distinguished from an algebraic norm $N : \mathcal{M} \to \mathbb{N}_0$:

(1) A *multiplicative projection* satisfies $\pi(x \star y) = \pi(x) \cdot \pi(y)$ for all $x, y \in \mathcal{M}$.

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(2) A *state-level identity* requires merely $\pi(x) = \pi(y)$ for specific pairs $x, y \in \mathcal{M}$, without imposing homomorphism properties on the entirety of $\mathcal{M} \times \mathcal{M}$.

3. Factorization and Irreducibility in $\mathcal{M}$

To avoid equivocation, we explicitly distinguish between irreducibility and primality within non-commutative or extended model spaces $\mathcal{M}$.

Definition 3.1 (Irreducibility)**.** An element $c \in \mathcal{M} \setminus (\mathcal{M}^\times \cup \{0\})$ is *irreducible* if $c = u \star v$ implies that either $u \in \mathcal{M}^\times$ or $v \in \mathcal{M}^\times$, where $\mathcal{M}^\times$ denotes the set of units in $\mathcal{M}$.

Remark 3.2 (Composite Norms of Irreducible Elements). An element $c \in \mathcal{M}$ can be irreducible in $\mathcal{M}$ while its projected scalar $n = \pi(c)$ is composite in $\mathbb{N}$. A well-known classical analogue is found in the Gaussian integers $\mathbb{Z}[i]$, where a rational prime $p \equiv 3 \pmod 4$ remains irreducible in $\mathbb{Z}[i]$, yet its norm $N(p) = p^2$ is composite in $\mathbb{N}$.

Lemma 3.3 (Action of Units on Reducibility)**.** *Let $\mathcal{M}$ be an algebra with unit group $\mathcal{M}^\times$. For any $x \in \mathcal{M}$ and $u, v \in \mathcal{M}^\times$, x is reducible in $\mathcal{M}$ if and only if $u \star x \star v$ is reducible in $\mathcal{M}$.*

Proof. Suppose $x = y \star z$ with non-units $y, z \in \mathcal{M} \setminus \mathcal{M}^\times$. Then $u \star x \star v = (u \star y) \star (z \star v)$. Since $u, v \in \mathcal{M}^\times$, $u \star y \notin \mathcal{M}^\times$ and $z \star v \notin \mathcal{M}^\times$. Thus $u \star x \star v$ is reducible. The converse follows by applying u^{-1} and v^{-1}. □

Corollary 3.4. *Multiplication by units preserves reducibility and irreducibility. However, unit multiplication cannot be assumed to preserve a uniquely defined "number of prime factors" unless $\mathcal{M}$ is proven to be a Unique Factorization Domain (UFD) or a length-preserving Krull monoid.*

4. The Fundamental Safeguard Theorem

We now state the principal theorem resolving the multi-representation question.

Theorem 4.1 (Scalar Invariance vs. Model Equivalence)**.** *Let $\mathcal{M}$ be a model space, $\pi : \mathcal{M} \to \mathbb{N}$ a scalar projection, and $x = a \star b \star E_1$, $y = c \star E_2$ two elements in $\mathcal{M}$. Assume that:*

$$\pi(x) = \pi(y) = n \in \mathbb{N}. \tag{4}$$

Then:

(1) *The natural scalar n obeys the Fundamental Theorem of Arithmetic uniquely:*

$$n = \prod_{p \text{ prime}} p^{v_p(n)}. \tag{5}$$

(2) *The equality $\pi(x) = \pi(y)$ does **not** imply $x = y$, nor does it imply association $x \sim y$.*
(3) *The equality $\pi(x) = \pi(y)$ does **not** imply isomorphism of state-level factorizations:*

$$\mathrm{Fact}(x) = \mathrm{Fact}(y). \tag{6}$$

Proof. Assertion (1) follows directly from the fact that $n \in \mathbb{N}$ and $\mathbb{N}$ is a Unique Factorization Monoid under standard multiplication.

For assertions (2) and (3), it suffices to give a counterexample to the implication. Let $\mathcal{M} = \mathbb{Z}[i]$, $\pi(z) = N(z) = z\bar{z}$. Set $a = 5$, so $n = 25$. Consider $x = 5 \in \mathbb{Z}[i]$ (which is irreducible in $\mathbb{Z}[i]$ as $5 \equiv 1 \pmod 4$ decomposes as $(2+i)(2-i)$, hence $x = (2+i)(2-i)$ is composite) and $y = 3+4i$. We have $\pi(x) = N(5) = 25$ and $\pi(y) = N(3+4i) = 3^2 + 4^2 = 25$. Here $\pi(x) = \pi(y) = 25$. However, $5 \neq 3 + 4i$, and $5 \not\sim 3 + 4i$ because $5/(3+4i) = (15 - 20i)/25 \notin \mathbb{Z}[i]$. Furthermore, in $\mathbb{Z}[i]$, $5 = (2+i)(2-i)$ has two irreducible factors of norm 5, whereas $3 + 4i = (2+i)^2$ consists of a square of a single prime up to units. Thus $\mathrm{Fact}(x) \neq \mathrm{Fact}(y)$. □

Corollary 4.2 (Core Protection Formula)**.** *The fundamental boundary formula protecting the consistency of the model framework is given by:*

$$\boxed{\pi(x) = \pi(y) \quad \not\Rightarrow \quad x = y, \quad x \sim y \quad \textit{or} \quad \mathrm{Fact}(x) = \mathrm{Fact}(y)} \tag{7}$$

5. Conclusion

The existence of dual representations $a \star b \star E_1$ and $c \star E_2$ yielding identical scalar projections $\pi(x) = \pi(y) = n$ presents zero contradiction to the Fundamental Theorem of Arithmetic. The uniqueness of factorization is a property of the scalar target domain $\mathbb{N}$, whereas the structural multiplicity resides in the fiber $\pi^{-1}(n) \subseteq \mathcal{M}$. Model states represent different geometric or algebraic configurations whose projected invariant scalar coincides, without collapsing the distinct state-level factorizations into a single identity.

References

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